

AMENDMENT NO. 3 JUNE 2019
TO
IS 14272 : 2011 AUTOMOTIVE VEHICLES — TYPES — TERMINOLOGY
(First Revision)

(Page 4, clause 3.6) — Substitute the following for the existing clauses:

‘3.6 Category L – Motor vehicles with less than four wheels and quadricycle.

3.6.1L5 Category – Three-Wheeler — A three wheeled motor vehicle with maximum speed exceeding 25 km and engine capacity exceeding 25 cc, if fitted with a thermic engine, or motor power exceeding 0.25 kW, if fitted with electric motor. This vehicle is normally used for,

- a) carrying persons; or
- b) carrying goods.

Semi-trailer may be attached, and where,

- 1) handle bar or steering wheel may be fitted;
- 2) gross vehicle weight will be limited to 1 500 kg, subject to the conditions given in 4;
- 3) in the case of semi-trailers being attached to a three wheeled tractor, the gross combination weight will be limited to 2 500 kg subject to the conditions given in 4; and
- 4) weight of rechargeable energy storage system (REESS) in the case of battery operated three wheelers, shall not be taken into account for the limitation the GVW/ GCW and for the purpose of classification

3.6.1.1 L5M Category — Passenger carrier (auto-rickshaw) — A three-wheeler on account of its technical features intended to carry passengers.

3.6.1.2 L5N Category — Goods carrier — A three wheeler on account of its technical features intended to carry goods.

3.6.1.3 A three-wheeler may fall under the category of L5M [Passenger carrier (Auto rickshaw)] or L5N (Goods carriage) depending on whether the weight of persons including driver for whom seating arrangements are provided is more than or less than the weight of goods carried.

If the following conditions are satisfied, a three wheeler comes under the category of L5N (Goods carriage) and not L5M [(Passenger carrier (Auto-rickshaw))]:

- a) A separate load body or compartment is provided for carrying the goods;
- b) Number of seating positions excluding the driver is not more than three; and
- c) Weight of goods carried by the vehicle is more than weight of persons carried, as calculated by following formula:

$$P = (A + B \times 68) > B \times 68$$

where

P = technically permissible maximum laden weight (GVW), in kg;

A = vehicle weight in the Kerb weight condition (as defined in IS 9211) + 68, in kg; and

B = number of seating positions excluding the driver.

NOTE — In the case of electric vehicles, the weight of rechargeable energy storage system (REESS) is to be subtracted from the Kerb weight.

Price Group 3

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(Page 4, clause 3.8.1) — Insert the following at the end of the clause:

‘Category A comprises all wheeled tractors; each wheeled tractor category described in points 3.8.1.1 to 3.8.1.12 is supplemented at the end by an ‘a’ or ‘b’ index according to its design speed:

- a) ‘a’ for wheeled tractors with a maximum design speed less than or equal to 40 km/h.
- b) ‘b’ for wheeled tractors with a maximum design speed more than 40 km/h.’

(Page 4, clause 3.8.1) — Insert the following new clauses:

3.8.1.1 *Category A 1*, comprises wheeled agricultural tractors, with the closest axle to the driver having a minimum track width of not less than 1 150 mm, with an unladen mass, in running order, of more than 600 kg, and with a ground clearance of not more than 1 000 mm.

3.8.1.2 *Category A 2*, comprises wheeled agricultural tractors with a minimum track width of less than 1 150 mm, with an unladen mass, in running order, of more than 600 kg, with a ground clearance of not more than 600 mm.

3.8.1.3 *Category A 3*, comprises wheeled agricultural tractors with an unladen mass, in running order, of not more than 600 kg.

3.8.1.4 *Category A 4*, comprises special purpose wheeled agricultural tractors.

3.8.1.4.1 *Category A 4.1 (high-clearance agricultural tractors)*, comprises tractors designed for working with high-growing crops, such as vines. They feature a raised chassis or section of chassis, enabling them to advance in parallel with the crop with left and right wheels on either side of one or more rows of the crop. They are intended for carrying or operating tools which may be fitted at the front, between the axles, at the rear or on a platform. When the tractor is in working position the ground clearance perpendicular to the crop rows exceeds 1 000 mm.

3.8.1.4.2 *Category A 4.2 (extra-wide agricultural tractors)*, comprises tractors characterized by their large dimensions, primarily intended for working large areas of farmland.

3.8.1.4.3 *Category A 4.3 (low-clearance agricultural tractors)*, comprises four-wheel drive tractors whose interchangeable equipment is intended for agricultural or forestry use and which are characterized by a supporting frame, equipped with one or more power take-offs, having a technically permissible mass no greater than 10 tonne, for which the ratio of this mass to the maximum unladen mass in running order is less than 2.5 and having the centre of gravity, measured in relation to the ground using the tyres normally fitted, of less than 850 mm.

3.8.1.5 *Hybrid agricultural tractor (A5)*, means a powered Agricultural Tractor equipped with at least two different energy converters, and two different energy storage systems (on-vehicle) for the purpose of vehicle propulsion.

3.8.1.6 *Alternate-fuel gas agricultural tractors (A 5.1)*, means a vehicle that is designed primarily for permanent running on LPG or NG/bio-methane or hydrogen, but may also have a petrol / Diesel system for emergency purposes or starting only, where the petrol tank does not contain more than 5 litre of petrol.

3.8.1.7 *Bi-fuel agricultural tractors (A 5.2)*, means a vehicle with two separate fuel storage systems that is designed to run on only one fuel at a time. The simultaneous use of both fuels is limited in amount and duration.

3.8.1.8 *Bi-fuel gas agricultural tractors (A 5.3)*, means a bi fuel vehicle that can run on petrol / diesel and also on either LPG, NG/bio-methane or hydrogen (gas mode).

3.8.1.9 *Alternative fuel agricultural tractors (A 5.4)*, means a vehicle designed to be capable of running on at least one type of fuel that is either gaseous at atmospheric temperature and pressure, or substantially non-mineral oil derived.

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3.8.1.10 Flex fuel agricultural tractors (A5.5), means a vehicle with one fuel storage system that can run on different mixtures of two or more fuels.

3.8.1.10.1 Flex fuel ethanol agricultural tractors (A 5.5.1), means a flex fuel vehicle that can run on petrol or a mixture of petrol and ethanol up to an 85/100 percent ethanol blend (E85/E100).

3.8.1.10.2 Flex fuel biodiesel agricultural tractors (A 5.5.2), means a flex fuel vehicle that can run on mineral diesel / petrol or a mixture of mineral diesel / petrol and bio-diesel.

3.8.1.11 Hybrid electric agricultural tractor (A 6), means an agricultural tractor that, for the purpose of mechanical propulsion, draws energy from both (a) and (b) of the following on-vehicle sources of stored energy/power:

- a) consumable fuel.
- b) a battery, capacitor, flywheel/generator or other electrical energy or power storage device.

This definition also includes agricultural tractor which draw energy from a consumable fuel only or Solar energy for the purpose of recharging the electrical energy/power storage device;

3.8.1.12 Pure electric agricultural tractor (A7), means an agricultural tractor powered by a system consisting of one or more electric energy including solar energy storage devices, one or more electric power conditioning devices and one or more electric machines that convert stored electric energy to mechanical energy delivered at the wheels for propulsion of the vehicle.’

[Page 4, clauses **3.10** to **3.10.2** (see also Amendment No. 2)] — Substitute the following for the existing clauses:

‘3.10 Category L7- Quadricycle — A vehicle as per Rule (2) of *Central Motor Vehicle Rules*, 1989.

3.10.1 Category L7-M – A quadricycle of category L7 used for carrying passengers, having seats not more than 4 (including driver) and kerb weight not exceeding 475 kg.

3.10.2 Category L7-N – A quadricycle of category L7 used for carrying goods, having seats not more than 2 (including driver) and kerb weight not exceeding 550 kg.

NOTE — The weight shall not include :

- a) Mass of rechargeable energy storage system (REESS) in the case of electric or hybrid vehicles; and
- b) Mass of gaseous fuel system including tanks for gaseous fuel storage in the case of mono-fuel, bi-fuel or multi-fuel vehicles.’

[Page 4, clause **3.10.2** (see also Amendment No. 2)] — Insert the following new clauses at the end of the clause:

‘3.11 Special Purpose Vehicle (SPV) — A vehicle of category L (only in case of road ambulance complying to AIS-125 (Part1)) M, N or T having specific technical features in order to perform a function which requires special arrangements and / or equipment.

3.12 Motor Caravan — A special purpose M category vehicle constructed to include living accommodation which contains at least the following equipment:

- a) Seats and table;
- b) Sleeping accommodation which may be converted from the seats;
- c) Cooking facilities; and
- d) Storage facilities.

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This equipment shall be rigidly fixed to the living compartment; however, the table may be designed to be easily removable

3.13 Road Ambulance — Road ambulance or ambulance is a specially equipped and ergonomically designed vehicle for transportation / emergent treatment of sick or injured people and capable of providing out of hospital medical care during transit / when stationary, commensurate with its designated level of care when appropriately staffed.

3.14 School Bus — Vehicles with a seating capacity of 13 passengers and above excluding driver designed and constructed specially used for transporting children to and from schools.

3.15 Sleeper Coach — Vehicles with a seating capacity of 13 passengers and above excluding driver designed and constructed with additional provision/berth for passengers to sleep while travelling.

3.16 Twinned Wheels — In the case of L category vehicles, two wheels mounted on the same axle, the distance between the centres of their areas of contact with the ground is equal to or less than 460 mm. Twinned wheels shall be considered as one wheel.

3.17 Battery Operated Vehicle (Two Wheeler) — A vehicle adapted for use upon roads and powered exclusively by an electric motor whose traction energy is supplied exclusively by traction battery installed in the vehicle.

Provided that a two wheeled battery operated vehicle shall not be deemed to be a motor vehicle if all the following conditions are verified and authorized by any testing agency specified in *Central Motor Vehicle Rule* 126, namely:

- a) Vehicle is equipped with an electric motor having thirty minute power less than 0.25 kW;
- b) Maximum speed of the vehicle is less than 25 km/h;
- c) Vehicle is fitted with suitable brakes and retro-reflective devices, that is one white reflector in the front and one red reflector at the rear;
- d) Unladen weight (excluding battery weight) of the vehicle is not more than 60 kg; and
- e) In case of pedal assisted vehicle equipped with an auxiliary electric motor, in addition to above, the thirty minute power of the motor is less than 0.25 kW, whose output is progressively reduced and finally cut off as the vehicle reaches a speed of 25 km/h, or sooner, if the cyclist stops pedaling.

3.18 E-rickshaw — A special purpose battery operated vehicle having three wheels and intended to provide last mile connectivity for transport of passengers for hire or reward, provided:

- a) Such vehicle is constructed or adapted to carry not more than four passengers, excluding the driver, and not more than forty kilograms luggage in total;
- b) Net power of its motor is not more than 2 000 W; and
- c) Maximum speed of the vehicle is not more than 25km/h.

3.19 E-cart — A special purpose battery operated vehicle having three wheels and intended to provide last mile connectivity for carrying goods for hire or reward, provided:

- a) Such vehicles is constructed or adapted for carrying goods by providing a separate load body or compartment with the maximum weight three hundred and ten kilograms in addition to driver;
- b) Net power of its motor is not more than 2 000 W; and
- c) Maximum speed of the vehicle is not more than 25 km/h.

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3.20 Combine Harvester — Means an agricultural equipment vehicle, self-propelled or agricultural tractor powered type (either coupled to the trailer for header assembly or any other attachment of the machine) designed to perform more than one of the following tasks namely:

- a)
 - 1) Picking,
 - 2) Harvesting ,
 - 3) Felling,
 - 4) Threshing,
 - 5) Separating,
 - 6) Cleaning,
 - 7) Debarking,
 - 8) Chopping,
 - 9) Billeting,
 - 10) Collecting, and
 - 11) Unloading crop, agricultural product such as agro - forestry product including but not limited to grain, sugarcane, cotton, fodder, straw or stalk, pulp wood, root and tuber crops like potato, onion, carrots while moving through the standing crop or agricultural produce.

and / or

- b) Arrangement of bagging with a pick-up attachment to use it for handling crop that has been swathed.

NOTE — Mounted / Semi-mounted / Trailed attachments are excluded from this definition.

3.21 Modular Hydraulic Trailer — A trailer module intended for carrying indivisible heavy or over-dimensional cargo and having the following features namely:

- a) Swing axles with hydraulic suspension;
- b) Independently steerable axles;
- c) Two or more axle rows;
- d) Suitable arrangement for joining such modules; longitudinally or laterally or both;
- e) Suitable provision for joining such separate modules with spacer beam arrangement or by bolster arrangement or by grinder bridge arrangement or by loading deck arrangement;
- f) Suitable drawbar arrangement for being pulled or pushed or self-propelled; and
- g) Fitted with suitable braking system.

3.22 Puller Tractor — A multi-axle haulage tractor of category N3 vehicle having:

- a) Suitable arrangement to pull or push modular hydraulic trailer or combination thereof under drawbar arrangement;
- b) Adequate ballast weight for providing traction;
- c) Minimum engine power of 260 hp and;
- d) Maximum speed not exceeding twenty five km/h while pulling load.

3.23 Power Tiller — Agricultural machinery used for soil preparation having a single axle in which the direction of travel and its control for field operation is performed by the operator walking behind it. This equipment may or may not have a riding attachment and when coupled to a trailer can be used for the transportation of goods. The maximum speed of the power tiller when coupled to a trailer shall not exceed 22 km/h. The maximum haulage capacity of the power tiller coupled to a trailer shall not exceed 1.5 tonne.

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3.24 Agricultural Trailer — A trailer generally left uncovered with single/ double axle construction which is coupled to an agricultural tractor by means of tow hooks and predominantly used for transporting agricultural materials.

3.25 Axle — The common axis of rotation of two or more wheels whether power driven or freely rotating, and whether in one or more segments located in the same plane perpendicular to the longitudinal centre-line of the vehicle, which transmits the load of the vehicle to the road through these wheels.

3.26 Group of Axle — An assemblage of two or more consecutive axles considered together in determining their combined load effect on a pavement structure.

3.27 Solo Axle — An axle that cannot be considered as part of a group of axles.

3.28 Tandem Axle — A group of two axles. Weight limits for tandem axle are as per requirements in *Central Motor Vehicle Rules*, 1989.

3.29 Tridem Axle — A group of three axles.

3.30 Airport Passenger Bus (Tarmac Bus) — A bus having doors on both sides which is exclusively used for the carrying the passengers from airport terminal to and from the aircraft and plying on the air side on the airport- terminal with maximum speed not exceeding thirty kilometer per hour.

3.31 Double Deck Vehicle — A vehicle where the provided spaces for passengers are arranged, at least in one part, in two superimposed levels and spaces for standing passengers are not provided in the upper deck.

3.32 Double Deck Articulated Vehicle — A vehicle which consists of two or more rigid sections which articulate relative to one another; the passenger compartments of each section intercommunicate on at least one deck so that passengers can move freely between them; the rigid sections are permanently connected so that they can only be separated by an operation involving facilities which are normally only found in a workshop.

3.33 A Tipper — N-category vehicle intended for carrying loose unpackaged material and provided with powered equipment for inclining the payload carrying area to deposit the payload on the ground.'